

Introduction

The work presented in this thesis was carried out in the cold-atoms laboratory of Professors Francesco Minardi and Marco Prevedelli at the Department of Physics and Astronomy “A. Righi” of the University of Bologna. The experiment aims to prepare a cold rubidium-87 (^{87}Rb) gas and load it into a Hollow-Core Photonic Crystal Fiber (HCPCF).

HCPCFs enable the confinement of atoms within the hollow core and their interaction with guided optical fields over extended propagation distances [1]. Such systems are relevant to matter-wave interferometry [2], quantum memories, precision measurements, and the study of collective phenomena [3, 1].

The experimental sequence consists of collecting the atoms in a Magneto-Optical Trap (MOT), further cooling them through optical molasses, and loading them into a 1064 nm Optical Dipole Trap (ODT). The atoms will subsequently be transported towards the HCPCF by means of an optical conveyor belt. The conveyor belt is based on two counter-propagating 1064 nm beams whose interference produces a one-dimensional optical lattice. A controlled frequency difference between the two beams sets the lattice in motion, enabling the transport of the trapped atoms.

This thesis focuses on the optimization of the optical molasses, the characterization of atom loading into the ODT, and the implementation of the second optical branch required for the future realization of the conveyor belt. In parallel, a multilevel numerical model of Electromagnetically Induced Transparency (EIT) cooling is developed to study the motional dynamics of the atoms and the possibility of cooling them after transport.

Chapter 1 introduces the theoretical background concerning HCPCFs, laser cooling, optical trapping, optical transport, and EIT cooling. Chapter 2 describes the experimental apparatus, the calibration and measurement procedures, the implementation of the second optical branch, the fiber-injection system, and the numerical model. Chapter 3 presents the experimental results on molasses optimization, ODT loading, and field stability, together with the simulated EIT-cooling dynamics.

The work presented here contributes to the development of the optical transport system and provides a numerical framework for the future implementation of EIT cooling in the experiment.